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Study of multi-dimensional network energy efficiency metrics

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Foreword

This Technical Report (TR) has been produced by ETSI Technical Committee Environmental Engineering (EE).

The present document was developed jointly by ETSI TC EE and ITU-T Study Group 5 and published respectively by ITU and ETSI as Supplement ITU-T L.Suppl. 62 [i.6] and ETSI Technical Report ETSI TR 104 084 (the present document), which are technically equivalent.

Modal verbs terminology

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Executive summary

Existing network energy efficiency metrics are typically only based on the amount of service provided, e.g. transported data volume, coverage area, or number of connections. Other dimensions of useful work, e.g. the quality of the service provided, is seldom reflected, and it is also not always clear if/how these metrics can be combined to give a more complete picture of the energy efficiency of a network. The scope of the present document is to study existing network energy efficiency metrics and outline proposals on how network energy efficiency metrics capturing multiple performance dimensions (service volume, service quality, service availability, etc.) can be formulated. These can be combinations of existing network energy efficiency metrics, but also new proposals capturing performance dimensions not considered today. The focus has been on network energy efficiency metrics to be used for monitoring and/or optimization of live networks, even though metrics for reporting and related implications have been also covered.

The analysis concludes that it is desirable that in addition to the existing mostly service-volume based EE metrics, also service quality is added into the EE assessment. In particular, service quality based on (average) user throughput may be valuable to consider in addition to data volume when assessing mobile network data energy efficiency, something that has also been concluded by 3GPP. It is also desirable to extend the current mobile network coverage energy efficiency analysis to consider differences in coverage quality in addition to coverage availability. Whether also service availability should be considered is less obvious, as the analysis conducted shows limited value to add such metrics into the assessment.

Three multi-dimensional metric proposals, i.e. structures for considering multiple performance dimensions in the EE assessment, have been defined and assessed in the present document:

- **Metric based on adjustment factor(s):** A base metric, typically the existing service volume-based EE metrics already defined, is extended (i.e. multiplied) by a dimension-less adjustment factor. The idea is that the adjustment factor captures another performance dimension and scales the base metric accordingly.
- **Multiplicative metric:** In this case, two performance indicators are multiplied and considered in relation to the network energy consumption.
- **Separate metrics per performance dimension:** Multiple single-dimensional energy efficiency metrics represent the different performance dimensions in the network energy efficiency evaluation.

Of these, the second proposal - multiplicative metric - has been assessed as least attractive. It can be difficult to understand, as it mixes units of different performance metrics and relates them to the energy consumption at the same time. It may then be difficult to understand how the metric behaves, and what it really tells. The other two have their pros and cons depending on the use case. Using separate metrics per performance dimension is beneficial for a more nuanced assessment, as it reveals all details of what is driving the results. However, it may make an optimization algorithm more complex with more variables to consider. A metric based on adjustment factors may hide nuances, but with just one or two performance indicators reflected this risk is reduced. Furthermore, if it is constructed in a sound way it should be easy to understand, e.g. with service volume scaled by a factor based on service quality. It may in particular be useful for network optimization.

For reporting, it is recommended to keep the existing single dimensional metrics, e.g. mobile network data energy efficiency, as they have been used for long time, are well-known, and also requested in some reporting frameworks.

Furthermore, the analysis also confirms that contextual factors such as geographical conditions, population density, topology, climate zones, and also regulatory obligations on coverage and service quality, may have a significant impact on the energy efficiency assessment and its' results. Therefore, this again emphasizes that the network level energy efficiency analysis and metrics should not be used for benchmarking of different networks (unless the contextual factors are the same or compensated for), rather the use case is for assessing improvements over time in a specific network and optimization of the network.

Introduction

A mobile network needs to fulfil several service requirements, such as being able to deliver a certain service volume with a given service quality and service availability. A number of different energy efficiency metrics have been defined throughout the years, e.g. for mobile networks in ETSI ES 203 228 [i.1]. These metrics are typically used to assess the energy efficiency of the network over time, but can also be used in network optimization algorithms for improving the network energy efficiency in an automated manner. The metrics should not be used for benchmarking of different networks, as contextual factors of different networks such as regulatory obligations, emergency service requirements, geographical and population characteristics, etc. significantly influence the metrics.

These existing network energy efficiency metrics are typically only based on the amount of service provided, e.g. transported data volume, coverage area, or number of connections. However, the quality of the service provided is seldom reflected, and it is also not always clear if/how these metrics can be combined to give a more complete picture of the energy efficiency of a network.

The present document studies existing network energy efficiency metrics and outlines proposals and analysis on how mobile network energy efficiency metrics capturing multiple performance dimensions (service volume, service quality, service availability, etc.) can be formulated.

1 Scope

The present document studies existing network energy efficiency metrics and outlines proposals how network energy efficiency metrics capturing multiple performance dimensions (service volume, service quality, service availability, etc.) could be formulated. These can be combinations of existing energy efficiency metrics, but also new proposals capturing performance dimensions not considered today. The focus is on network energy efficiency metrics to be used for monitoring and/or optimization of live networks.

2 References

2.1 Normative references

Normative references are not applicable in the present document.

2.2 Informative references

References are either specific (identified by date of publication and/or edition number or version number) or non-specific. For specific references, only the cited version applies. For non-specific references, the latest version of the referenced document (including any amendments) applies.

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The following referenced documents may be useful in implementing an ETSI deliverable or add to the reader's understanding, but are not required for conformance to the present document.

[i.1] ETSI ES 203 228 (V1.4.1): "Environmental Engineering (EE); Assessment of mobile network energy efficiency".

[i.2] ETSI TS 128 554 (V19.5.0): "5G; Management and orchestration; 5G end to end Key Performance Indicators (KPI) (3GPP TS 28.554 version 19.5.0 Release 19)".

[i.3] ETSI TR 128 913 (V18.0.1), "5G; Study on new aspects of Energy Efficiency (EE) for 5G phase 2 (3GPP TR 28.913 version 18.0.1 Release 18)".

[i.4] ETSI TS 132 450 (V18.0.0): "Universal Mobile Telecommunications System (UMTS); LTE; Telecommunication management; Key Performance Indicators (KPI) for Evolved Universal Terrestrial Radio Access Network (E-UTRAN): Definitions (3GPP TS 32.450 version 18.0.0 Release 18)".

[i.5] 3GPP TR 28.880 (V19.0.0 - 2024-12): "Study on energy efficiency and energy saving aspects of 5G networks and services (Release 19)".

[i.6] ITU-T L.Suppl. 62: "Study of Multi-dimensional network energy efficiency metrics".

[i.7] Recommendation ITU-T L.1015 (05/19): "Criteria for evaluation of the environmental impact of mobile phones".

[i.8] Recommendation ITU-T K.114 (08/22): "Electromagnetic compatibility requirements and measurement methods for digital cellular mobile communication base station equipment".

[i.9] Recommendation ITU-T E.805 (12/19): "Strategies to establish quality regulatory frameworks".

[i.10] ETSI ES 202 336-12 (V1.3.1): "Environmental Engineering (EE); Monitoring and control interface for infrastructure equipment (power, cooling and building environment systems used in telecommunication networks); Part 12: ICT equipment power, energy and environmental parameters monitoring information model".

[i.11] JRC144975 (2026) Baldini, G. Cerutti I: "EU Code of Conduct for the sustainability of telecommunications networks".

3 Definition of terms, symbols and abbreviations

3.1 Terms

For the purposes of the present document, the following terms apply:

end-to-end latency: time taken to transfer a given piece of information from a source to a destination, measured at the communication interface, from the moment it is transmitted by the source to the moment it is successfully received at the destination

NOTE: As defined in [i.1].

energy efficiency: relation between the useful output and energy/power consumption

NOTE: As defined in [i.1].

multi-dimensional network energy efficiency: network energy efficiency assessment or metric based on multiple performance indicators from different performance dimensions

reliability: probability that a product functions as required under given conditions, including maintenance, for a given duration without failure

NOTE: As defined in [i.7].

service availability: availability of the service provided by the network, for example in time or space

service quality: quality of the telecommunication services provided by an operator

NOTE: As defined in [i.9].

service volume: volume of the service provided by the network, for example data volume, coverage area, or number of connections

throughput: number of payload bits successfully received per second for a reference measurement channel under a specified reference condition

NOTE: As defined in [i.8].

3.2 Symbols

Void.

3.3 Abbreviations

For the purposes of the present document, the following abbreviations apply:

2G	Second Generation
3G	Third Generation
3GPP	3rd Generation Partnership Project
4G	Fourth Generation
5G	Fifth Generation
B2B	Business-to-Business
B2C	Business-to-Consumer
BS	Base Station
CO2e	Carbon Dioxide equivalent
CoA	Coverage Area
CS	Circuit Switched
CSP	Communication Service Provider
DL	DownLink
DV	Data Volume
EC	Energy Consumption

EE	Energy Efficiency
eMBB	enhanced Mobile Broadband
eNB	evolved Node B
FDD	Frequency Division Duplex
FWA	Fixed Wireless Access
gNB	Next Generation Node B
HW	Hardware
ICT	Information and Communication Technologies
IoT	Internet of Things
IP	Internet Protocol
KPI	Key Performance Indicator
LTE	Long Term Evolution
MDEE	Multi-Dimensional Energy Efficiency
mMTC	massive Machine Type Communications
MN	Mobile Network
MNO	Mobile Network Operator
MTC	Machine Type Communication
NG-RAN	New Generation Radio Access Network
NR	New Radio
NSA	Non-Stand Alone
NTN	Non-Terrestrial Networks
O&M	Operation and Maintenance
OPEX	OPerational EXpenditures
O-RAN	Open RAN
OTT	Over The Top
PDCP	Packet Data Convergence Protocol
QCI	QoS Class Identifier
QoS	Quality of Service
RAB	Radio Access Bearer
RAN	Radio Access Network
RAT	Radio Access Technology
RF	Radio Frequency
RLC	Radio Link Control
RRC	Radio Resource Control
RSRP	Reference Signal Received Power
SA	Stand-Alone
SDU	Service Data Unit
TTI	Transmission Time Interval
UE	User Equipment
UL	Uplink
uRLLC	ultra-Reliable Low Latency Communication
VoLTE	Voice over LTE
VoNR	Voice over NR

4 Multi-dimensional network energy efficiency

A Mobile Network (MN) needs to fulfill several service requirements. It need to provide reliable coverage where the users want to access service, it need to provide enough capacity so that the traffic demand can be served, and it need to provide a service quality such that the users are satisfied with their experience of the service. Mobile Network Operators (MNOs) are continuously investing in their networks to make sure these service requirements, which evolve over time, are fulfilled. These investments are typically materialized in equipment consuming energy, and energy efficiency is an important objective. Energy efficiency is often defined as the useful output of a product or service divided by the energy consumption of the product or the service:

$$Energy\ Efficiency = \frac{Useful\ work}{Energy\ consumption}$$

A number of different energy efficiency metrics have been defined throughout the years, e.g. for mobile networks in ETSI ES 203 228 [i.1]. These network energy efficiency metrics are typically based on the amount of service provided, e.g. transported data volume, coverage area, or number of connections.

However, a drawback with these metrics is that they only consider the amount of service, but not the quality of the service provided which also is an important useful output of the network. In fact, mobile networks today may have similar data volume energy efficiency, but the user quality may vary greatly.

With the continuous development of networks, new services and new scenarios emerge. The focus of services varies in different phases and scenarios of network development, both in terms of increasing traffic demand to service, and the requirements for service quality. To allow for a more comprehensive energy efficiency evaluation as well as optimization in different network development phases and scenarios, it is desirable that also other performance dimensions than the service volume is reflected, e.g. the service quality.

5 Existing network energy efficiency metrics

ETSI ES 203 228 [i.1] defines mobile network energy consumption metrics and mobile network energy efficiency metrics based on a number of performance metrics coupled to use case families.

For the enhanced Mobile Broadband (eMBB) use case family, the energy efficiency metrics Mobile Network data Energy Efficiency ($EE_{MN,DV}$) and Mobile Network coverage Energy Efficiency ($EE_{MN,CoA}$) measured in bit/J and m^2/J, respectively, are defined. They are derived from performance metrics of the Mobile Network (MN) under investigation, in these cases the total data volume (DV_{MN}) delivered by all its equipment and its global coverage area (CoA_{MN}), respectively, and divided by the mobile network energy consumption (EC_{MN}):

$$EE_{MN,DV} = \frac{DV_{MN}}{EC_{MN}}$$

$$EE_{MN,CoA} = \frac{\mathrm{CoA_{MN}}}{EC_{MN}}$$

Mobile Network coverage Energy Efficiency ($EE_{MN,CoA}$) is mainly used to complement Mobile Network data Energy Efficiency ($EE_{MN,DV}$) for mobile networks handling low data volumes, in particular in rural or deep rural areas. It is worth noting, that $EE_{MN,CoA}$ is the only one of the defined mobile network energy efficiency metrics that involves a quality factor. In ETSI ES 203 228 [i.1], the coverage quality factor (CoA_Q) measures the performances of the network within the actually covered fraction of the planned total coverage area. UE reports such as failed call attempts are used to determine how well the users within the coverage area are covered, and the coverage quality factor is on a high level defined as:

Q = 1 - "percentage of users/sessions with coverage failure"

For the ultra-Reliable Low Latency Communications (uRLLC) use case family, ETSI ES 203 228 [i.1] defines a mobile network energy efficiency metric based on the end-to-end latency of the mobile network ($T_{e2e,MN}$). The mobile network latency energy efficiency ($EE_{MN,L}$) is defined as the inverse ratio of the end-to-end user plane latency and the energy consumed by the MN:

$$EE_{MN,L} = \frac{1}{T_{e2e,MN} * EC_{MN}}$$

where $EE_{MN,L}$ is expressed in ms^{-1}/J.

For the massive Machine Type Communications (mMTC) use cases, ETSI ES 203 228 [i.1] defines a mobile network energy efficiency metric based on the number of subscribers (N_{MMTC}) registered to the network. In this case, the EE metric is defined as follows:

$$EE_{MN,MMTC} = \frac{N_{MMTC}}{EC_{MN}}$$

For more detailed information on how these mobile network energy efficiency metrics are calculated, and how the mobile network energy consumption (EC_{MN}) as well as the respective performance metrics are measured, see ETSI ES 203 228 [i.1].

6 Performance dimensions

6.1 Introduction

A mobile network needs to provide several services, and fulfil many requirements. For long time mobile networks have been used for voice communication, then for consuming multimedia services, and with the introduction of 5G more and more of machine type communications. All of these communication forms are characterized by different Key Performance Indicators (KPIs), telling how well the network is delivering its services.

The performance indicators can typically be divided into different classes, or dimensions. The three common performance dimensions are the following:

- *Service volume*, typically measuring the volume or amount of service provided. Examples can be the data volume delivered, the area covered, the number of supported connections, etc.
- *Service quality*, measuring the quality of the service provided. Examples are user throughput, coverage quality or connection quality.
- *Service availability*, typically measuring the availability of the service. An obvious example in a wireless network is the area coverage of the service, but one can also envision other KPIs belonging to this performance dimension such as service availability in time.

One can probably formulate more performance dimensions, but these are a good starting point in order to keep the complexity down. Furthermore, as can be seen of the examples, some KPIs may fall into more than one of the performance dimensions depending on the type of service and how important the different KPIs are. Therefore, how to classify different KPIs typically depends on type of service or use case. This is in line with how energy efficiency KPIs have been defined in ETSI ES 203 228 [i.1], as they are typically based on the most important performance KPI for the use case.

To exemplify this, the three 5G use case families eMBB, mMTC and uRLLC will be examined, and KPIs belonging to the three performance dimensions will be identified for each of the use case families.

6.2 Service volume

6.2.1 Enhanced Mobile Broadband (eMBB)

Mobile broadband is still the absolutely dominating use case for mobile networks, and is expected to continue be so even though other use cases are emerging today. As can be seen from the existing eMBB energy efficiency metrics, two important performance metrics for eMBB are the total data volume (DV_{MN}) delivered and the coverage area (CoA_{MN}), respectively. These can be seen as belonging to the performance dimension service volume, but representing two different sub use cases; capacity sites and coverage sites.

6.2.2 Massive Machine Type Communications (mMTC)

For the mMTC use case family, an important performance indicator is the number of connected users, or as already used in ETSI ES 203 228 [i.1] the number of subscribers ($N_{MMTC)}$ registered to the network. This is clearly a service volume performance indicator.

6.2.3 Ultra-Reliable Low-Latency Communications (uRLLC)

The uRLLC use case family covers many different use cases. In some use cases it is the reliability that is the most important performance matric, while in others it is low latency that is the most important metric, and in some both are equally important. Furthermore, there may also be use cases where data volume is an important performance metric in addition to the reliability and low latency metrics.

So far, ETSI ES 203 228 [i.1] has defined the uRLLC energy efficiency metric based on the latency metric (or rather the inverse of it), ($1/T_{e2e,MN}$). It can therefore be seen as a service volume performance indicator (the lower latency, the better).

However, 3GPP has in [i.2] defined an uRLLC energy efficiency metric based on both latency and data volume, (DV_{MN} /$T_{e2e,MN}$). This KPI is applicable for the cases where, for example, the uRLLC is deployed and operators want to evaluate the Energy Efficiency at different periods of time, such as the busy hours in the morning and the idle hours in the mid night, in which both latency performance and the data volume performance can vary.

6.3 Service quality

6.3.1 Enhanced Mobile Broadband (eMBB)

Already present in ETSI ES 203 228 [i.1], is the coverage quality factor (*CoA_Q*), which adds a service quality performance dimension to the eMBB sub use case coverage sites. However, as currently defined the coverage quality factor is more a service availability indicator, and an alternative definition may be needed. For capacity sites, where the service volume is measured in data volume, a commonly used service quality indicator is the user throughput, which can be measured both in DL and UL and considered separately or jointly.

6.3.2 Massive Machine Type Communications (mMTC)

For mMTC, connection quality is a natural service quality indicator.

6.3.3 Ultra-Reliable Low-Latency Communications (uRLLC)

Performance of uRLLC is in addition to low latency also reflected by its reliability. The lower the latency or higher the reliability of uRLLC, the higher its performance is. 3GPP discuss this in [i.3], that an operator may want to assess energy efficiency with respect to reliability of uRLLC, e.g. for communication services requiring very high reliability, such as Cloud/Edge/Split Rendering, Gaming or Interactive Data Exchanging, wireless road-side infrastructure backhaul, etc. Reliability is a very important performance metric for such use cases. Reliability can be considered to be a service quality indicator.

6.4 Service availability

6.4.1 Enhanced Mobile Broadband (eMBB)

The straight-forward indicator of service availability in an eMBB scenario would be the coverage, e.g. the coverage area where a targeted service quality can be maintained or guaranteed. At the same time, the coverage area is already used as the service volume indicator for coverage sites, which need to be considered in the continued analysis. One way forward could be to use a coverage quality indicator (*CoA_Q*) as service availability metric, as mentioned in clause 6.3.1 above. Further availability indicators could be cell availability or RAN availability, which provides the average time availability duration per cell or the entire RAN, respectively [i.2].

6.4.2 Massive Machine Type Communications (mMTC)

For mMTC, the same service availability performance indicators as for eMBB can be considered.

6.4.3 Ultra-Reliable Low-Latency Communications (uRLLC)

Also here, the same service availability performance indicators as for eMBB can be considered.

6.5 Summary

Table 1 summarizes the key performance indicators, mapped to performance dimensions, for the different use case families.

Table 1: Main performance indicators for use case families

Use case family	Main performance indicators, divided on performance dimensions		
	Service volume	**Service quality**	**Service availability**
Enhanced mobile broadband (eMBB)	Data volume (capacity sites) Coverage area (coverage sites)	User throughput (capacity sites) Coverage quality (coverage sites)	Coverage (area or quality) alternatively Cell or RAN availability
Massive machine type communications (mMTC)	Number of connections or subscribers	Connection quality	See eMBB
Ultra-reliable low latency communications (uRLLC)	Data volume 1/Latency	Reliability	See eMBB

7 Multi-dimensional network energy efficiency metric proposals

7.1 General formulation

To allow for a more comprehensive network energy efficiency evaluation, and to better reflect the full performance of the network in the energy efficiency metric, the network energy efficiency metrics should take multiple performance dimensions into account. A general formulation, based on the high-level definition of energy efficiency in clause 4 above, can be as follows.

$$Energy\ Efficiency = \frac{f\ (service\ volume, service\ quality, service\ availability, \dots)}{Energy\ consumption}$$

This formulation indicates that the numerator should be a function of the service volume, service quality, service availability (and potential other performance dimensions), hence reflecting several performance dimensions. However, this general formulation should not be interpreted strictly mathematical, but rather reflecting the philosophy to consider several performance dimensions when assessing energy efficiency.

There are several reasons for why this could be beneficial. One is that 5G networks, not only target at carrying more data volume and providing a better coverage, but also at providing differentiation via Quality of Service (QoS). The current volume-based network energy efficiency metrics fail to capture MNOs' efforts to provide such QoS differentiation. Another is network optimization, where energy saving algorithms in certain cases may be misled by the volume-based energy efficiency metrics. Having the quality dimension reflected in the network energy efficiency metrics would solve many of these issues.

The service function in the numerator can be defined in different ways, and the following sections will outline some alternatives with their pros and cons.

7.2 Metric based on adjustment factor(s)

7.2.1 General

One straight-forward approach to formulate the multi-dimensional metric is to take the existing volume-based metrics as starting point, and to introduce adjustment factors to reflect the other performance dimensions. The adjustment factors would be dimensionless, in order not to affect the unit of the metric.

As an example, it is here illustrated how such an adjusted energy efficiency metric could be formulated for the eMBB use case family, i.e. how to adjust the network data energy efficiency metric to take service quality into account:

$$Energy\ Efficiency = \frac{Data\ volume}{Energy\ consumption} \times Factor_{quality}$$

The adjustment factor, in this case $Factor_{quality}$, should represent the service quality. In this simple example, it is defined as the average downlink user throughput in relation to the targeted downlink user throughput that the MNO wants to offer to its customers:

$$Factor_{quality} = \frac{Average\ downlink\ user\ throughput}{Average\ target\ downlink\ user\ throughput}$$

Hence, it describes how well the network is able to deliver the intended service quality, in this case, downlink user throughput. The average downlink throughput is easy to calculate based on standardized counters available in the operation and maintenance (O&M) system (see [i.4] and [i.2]), which is an advantage. For details about this, see Annex A.

As an example, take the networks introduced in Table 2 and assume that the average targeted downlink user throughput is 10 Mbit/s. Then the adjustment factors (in this case, $Factor_{quality}$) and the adjusted energy efficiencies for the different networks can be calculated. The results are summarized in Table 2. It can be seen that network A, which is the second best of the three in terms of network data energy efficiency, but is rather poor in delivering the targeted downlink user throughput of 10 Mbit/s, becomes the worst of the three in terms of QoS-adjusted energy efficiency. Network B, which is slightly worse that network A in terms of network data energy efficiency, but good in delivering the targeted downlink user throughput, is better than network A when the QoS-adjusted energy efficiencies are compared. Finally, network C is the best both before and after QoS adjustment. However, note that its delivered downlink user throughput is much higher than the targeted downlink user throughput. In order not to benefit from this over-provisioning, the adjustment factor has been limited to 1.

Table 2: Comparison of network data energy efficiency, user throughput, and QoS-adjusted energy efficiency in different networks

Network	Network Data Energy Efficiency (GB/kWh)	Average DL User Throughput (Mbit/s)	$Factor_{quality}$	QoS-adjusted Network Data Energy Efficiency (GB/kWh)
A	6,4	6,3	0,63	4,0
B	6,2	9,6	0,96	6,0
C	6,7	20,1	1	6,7

Similarly, the network coverage energy efficiency metric can be adjusted to take service quality into account:

$$Energy\ Efficiency = \frac{Coverage\ area}{Energy\ consumption} \times Factor_{quality}$$

In this case, the adjustment factor, $Factor_{quality}$, should represent the coverage quality. As mentioned earlier, there is already a definition of coverage quality provided in ETSI ES 203 228 [i.1]. However, that definition is focused on the limit where the service drops, and is already used when calculating the coverage area. Hence, it can be seen as an indication of the service availability. What is needed here, is a comparative measure of the quality of the coverage. A further discussion on this difference can be found in Annex B.

To design this comparative measure of the quality of the coverage, the cell edge user throughput in relation to the targeted cell edge user throughput is used:

$$Factor_{quality} = \frac{Cell\ edge\ user\ throughput}{Target\ cell\ edge\ user\ throughput}$$

Hence, it describes how well the network is able to deliver the intended coverage quality, in this case a certain cell edge user throughput. However, to measure cell edge user throughput in a live network is not straight-forward, and a better approach could be to use RSRP reports to represent the coverage quality (e.g. the amount of "good" RSRP reports in relation to the total amount of RSRP reports). Further details on this can be found in clause A.3, however, throughout the present clause the cell edge throughput is used for illustrative purposes.

So far, the examples presented has taken one additional performance dimension into account in the adjustment factor, but it is straight-forward to generalize the metric based on adjustment factors to reflect more performance dimensions. For example, the network data energy efficiency metric can be generalized to take both user throughput and coverage quality into account by defining the adjustment factor, $Factor_{quality}$, as:

$$Factor_{quality} = Factor_{throughput} \times Factor_{coverage} =$$

$$\frac{Average\ downlink\ user\ throughput}{Average\ target\ downlink\ user\ throughput} \times \frac{Cell\ edge\ user\ throughput}{Target\ cell\ edge\ user\ throughput}$$

As an example, Table 3 illustrates how the network data energy efficiency can be adjusted based on both user throughput and coverage quality for three different networks. The targeted user throughput in this example is 10 Mbit/s and the targeted cell edge user throughput is 1 Mbit/s.

Note that networks A and B have average user throughput and cell edge user throughput above the target values, and in order to not benefit from this overprovisioning the adjustment factors are limited to 1. Network C, on the other hand, who originally has the best network data energy efficiency of 8 GB/kWh, delivers both average user throughput and cell edge throughput below the targeted values. This results in adjustment factors < 1, and the adjusted network data energy efficiency becomes 6,5 GB/kWh.

Table 3: Comparison of network data energy efficiency, user throughput, coverage quality and adjusted energy efficiency in different networks

Network	Network Data Energy Efficiency (GB/kWh)	Average User Throughput (Mbit/s)	$Factor_{tp}$	Cell Edge User Throughput (Mbit/s)	$Factor_{cov.q}$	Adjusted Network Data Energy Efficiency (GB/kWh)
A	5	13	1	1,2	1	5 × 1 × 1 = 5
B	7	12	1	1,1	1	7 × 1 × 1 = 7
C	8	9	0,9	0,9	0,9	8 × 0,9 × 0,9 = 6,5

Finally, in the most generalized form, one can also introduce weighting factors for the different performance dimensions used in the adjustment factor. However, exactly how to do this need further study. Two options are:

$$Factor_{quality} = \prod_{i=1}^{n} F_i^{w_i}$$

or

$$Factor_{quality} = \sum_{i=1}^{n} w_i \times F_i$$

where F_i is the adjustment factor of different performance dimensions, e.g. user throughput, coverage quality, etc., and w_i is the weighting factor of the different performance dimensions, which can be flexibly defined for different scenarios. Note that these two options provide different results, and it need to be further studied which one is to be preferred when more than one performance dimension is used for adjustment.

Similar multi-dimensional metric formulations based on adjustment factor(s) can be obtained for the mMTC and uRLLC use case families.

7.2.2 Limitation of adjustment factors

In the examples in Table 2 and Table 3, the adjustment factors have been limited to 1. In other words, when the adjustment factor is calculated to be larger than 1, which means that the measured quality is higher than the targeted quality, the adjustment factor has been limited to 1. In the case for DL user throughput, this means the following formula is used:

$$Factor_{quality} = \min\left(1, \frac{Average\ measured\ downlink\ user\ throughput}{Average\ target\ downlink\ user\ throughput}\right)$$

Depending on the use case for the network energy efficiency metric, other options could also be considered. The formulation above is suitable when assessing and comparing different cases as one could argue that there should be no benefit from what could be seen as an over-dimensioning. In fact, this could even be taken one step further by inverting the adjustment factor when it becomes larger than 1. The motivation would be that since more resources than necessary have been used in the network (to deliver better quality than targeted), the network is operating with non-optimal resource usage.

However, there could also be use cases for the network energy efficiency metric where such limitation is not desirable. For example, with network optimization algorithms that activate, de-activate, and also re-distribute resources in the network it is necessary to have the full picture of what happens in the network, also when and where quality is over-provisioned, in order to take the correct decisions regarding de-activation and/or re-distribution of resources. An example is given in Annex C.

Hence, whether limitation of the adjustment factors should be used or not, need to be considered depending on the use case of the network energy efficiency metric.

7.3 Multiplicative metric

Another option to formulate a multi-dimensional metric is to formulate the numerator of the general energy efficiency definition in clause 7.1 as a multiplication of the performance dimensions. An example of such a metric is mentioned in clause 6.2.3 above, 3GPP has in ETSI TS 128 554 [i.2] defined an uRLLC energy efficiency metric based on both latency and data volume, (DV_{MN} /$T_{e2e,MN}$). The energy efficiency metric is in this case formulated as:

$$EE_{MN,DV,L} = \frac{DV_{MN}}{T_{e2e,MN} * EC_{MN}}$$

where $EE_{MN,DV,L}$ is expressed in bits/s*J. This KPI is applicable for the cases where, for example, the uRLLC is deployed and operators want to evaluate the Energy Efficiency at different periods of time, such as the busy hours in the morning and the idle hours in the mid night, in which both latency performance and the data volume performance can vary.

7.4 Separate metrics per performance dimension

A third approach could be to not try to capture multiple performance dimensions in a single energy efficiency metric, but instead let multiple single-dimensional energy efficiency metrics represent the different performance dimensions in the network energy efficiency evaluation.

Similar to the present document, 3GPP has in [i.5] identified network data traffic, network quality, and network availability as performance dimensions. These correspond to what in the present document is referred to as service volume, service quality, and service availability.

For the network data traffic dimension, 3GPP propose to use the already existing energy efficiency metrics based on data volume (for eMBB) and the number of connections (for mMTC). In ETSI, that would correspond to the already existing energy efficiency metrics in ETSI ES 203 228 [i.1], that are based on service volume. These metrics are also introduced in clause 5 in the present document.

When assessing network energy efficiency from the network quality dimension, 3GPP propose in [i.5] to use the average DL user throughput and the average UL user throughput, respectively, to reflect the network quality dimension. However, note that the network quality performance includes also other aspects such as coverage quality, or connection quality, etc., even though here the focus is only on the user throughput aspect.

The resulting new EE KPIs, which are evaluated as the ratio between performance (i.e. user throughput KPIs) and energy consumption, are captured in Table 4.

Table 4: New EE KPIs evaluated from network quality performance dimension

Performance dimension to evaluate EE KPIs		New EE KPIs	Note
Network quality	**UE throughput** as the network quality performance indicator	$\frac{DL\ RAN\ UE\ throughput}{EC_{per\ gNB}}$	The unit of this KPI is kbit/(s*J) This new EE KPI is obtained by DL RAN UE throughput divided by the average energy consumption per gNB over the same observation period. The average energy consumption per gNB is obtained by summing up the Energy Consumption (EC_{gNB}) of all the gNBs and then dividing the result by the total number of gNBs. EC_{gNB} is defined in clause 6.7.3.4.2 of ETSI TS 128 554 [i.2].
		$\frac{UL\ RAN\ UE\ throughput}{EC_{per\ gNB}}$	The unit of this KPI is kbit/(s*J). This new EE KPI is obtained by UL RAN UE throughput divided by the average energy consumption per gNB over the same observation period. The average energy consumption per gNB is obtained by summing up the Energy Consumption (EC_{gNB}) of all the gNBs and then dividing the result by the total number of gNBs. EC_{gNB} is defined in clause 6.7.3.4.2 of ETSI TS 128 554 [i.2].

For the availability performance dimension, existing KPI:s in ETSI TS 128 554 [i.2] are cell availability and RAN availability, which provides the average availability time duration per cell or the entire RAN. Hence, it is possible to define new EE KPI:s for the availability dimension by dividing these KPI:s with energy consumption according to Table 5.

Table 5: New EE KPIs evaluated from network availability performance dimension

Performance dimension to evaluate EE KPIs		New EE KPIs	Note
Network availability	**RAN Availability** as the availability performance indicator	$\frac{Radio\ access\ network\ availability}{EC_{per\ cell}}$	The unit of this KPI is s/J. This new EE KPI is obtained by dividing Radio access network availability KPI by the average energy consumption per cell of the RAN over the same observation period. The average energy consumption per cell of the RAN is obtained by considering summation of energy consumption of all gNB/CU in the RAN for a given time period and dividing it by total number of cells in the RAN that are operational in same time period.. NG-RAN EC is defined in clause 6.7.3.4.1 of ETSI TS 128 554 [i.2].
	Cell Availability as the availability performance indicator	Cell availability divided by average energy consumption per cell in a gNB	The unit of this KPI is s/J. This new EE KPI is obtained by considering Cell Availability KPI in numerator and dividing it by the average energy consumption per cell in a gNB over the same observation period. The average energy consumption per cell in a gNB is obtained by considering energy consumption of gNB for a given time period and dividing it by total number of cells in a gNB that are operational in same time duration/period. EC_{gNB} is defined in clause 6.7.3.4.2 of ETSI TS 128 554 [i.2].

The new energy efficiency metrics in Table 4 and Table 5, together with the existing energy efficiency metrics in ETSI ES 203 228 [i.1], forms a collection of single-dimensional energy efficiency metrics reflecting the three performance dimensions service volume, service quality, and service availability. Together, they can provide a picture of multi-dimensional energy efficiency of a network.

8 Analysis of metric proposals

8.1 Use cases

8.1.1 Introduction

ETSI ES 203 228 [i.1] aims to provide a better understanding of the Energy Efficiency (EE) of mobile networks in particular considering the networks' evolution in different periods of time. The focus is on metrics for energy efficiency and methods of assessing (and measuring) energy efficiency in operational networks. There are several potential users of ETSI ES 203 228 [i.1], e.g. MNO/CSP (Mobile Network Operators/Communication Service Providers) (and indirectly the suppliers of their network equipment and analytics), telecom regulators, neutral hosts, tower companies, B2B consumers, B2C consumers, etc.

Therefore, in order to properly frame the discussion on the analysis of the potential changes to the metrics and methodology to consider multiple performance dimensions proposed in clause 7 above, it is relevant to consider some potential relevant mobile network EE assessment use cases.

8.1.2 Mobile Network EE based network optimization

Description: Trace, compare and optimize the EE of a mobile network or a sub-network thereof over time, considering its evolution or changes e.g. traffic, feature activation, HW modernization, RF band refarming, parameter/configuration optimization, etc.

Key drivers: OPEX reduction, CO2e emission reduction.

8.1.3 Mobile Network EE based service delivery optimization

Description: Trace, compare and optimize the EE of a mobile network or a sub-network thereof over time, considering the service(s) delivered to customers and subscribers.

Challenges: It can become more complicated when the MNO has many different types of customers/subscription/service offerings, and also with the transversal implementation of e.g. network/service slices, or coverage of special customers like the industrial sector.

Key drivers: Revenue/cost optimization (and customer satisfaction at lowest possible energy cost) as well as CO2e emission reduction, possibly per service delivered.

8.1.4 Mobile Network EE assessment between different types of mobile network architecture

Description: Track and compare EE of different architecture or major configuration of mobile networks or sub-networks thereof within live conditions, e.g. between different deployed clusters with for instance cloud RAN, O-RAN, etc.

Key drivers: Consider EE in term of impacts versus performances as another input criteria for network architecture decision.

8.1.5 EE monitoring and reporting

Description: Voluntary or mandatory reporting of energy efficiency metric(s), with a focus on delivering wireless connectivity, broadband and services, at low/lower energy/CO2e impacts.

Key drivers: Environmental impact assessment as well as possibly energy dependency for connectivity infrastructure and services.

8.2 Evaluation of new and/or revised EE metrics

Besides the context of the relevant use cases clarified in clause 8.1 above, the new or revised EE metrics listed in clause 7 above are evaluated from a number of evaluation criteria in Table 6.

Table 6: Evaluation of new and/or revised EE metrics (black font for neutral comments, blue for positive and red for issues)

EE Metric	Added value	Relevance to use cases	Ease of understanding	Ease of measurement	Reproducibility	Reliability & accuracy	Energy proportionality	Scalability in scope and time	Standalone, composite or both
Name, formula, standard or TR referenced clauses	Value added compared to existing EE metrics in ETSI ES 203 228 [i.1].	Whether and how the metric remains relevant and beneficial for the use cases listed in clause 8.1 or only some of them?	How easy is it to understand and communicate about the metric, how it behaves, what it means and represents?	What is the complexity level or overhead of measuring, collecting and computing the metric?	Its measurement methodology supports reproducibility i.e. under similar circumstances and conditions, the EE metric will give similar results.	Its measurement methodology is precise enough to deliver good representation of the reality with low uncertainty level.	The correlation level between the selected useful work and the energy consumption pattern.	Both the implementation feasibility and benefit of using the metric from whole network to component level (e.g. BS, cell) as well as for a full year to hourly based.	A standalone metric or as a multiplicative factor (or both)?
Service volume - Data Volume $EE_{MN,DV}$ [b/J] $= \frac{DV_{MN}}{EC_{MN}}$ the total data volume (DV_{MN}) delivered by all its equipment divided by the Energy Consumption (EC_{MN}) of all its equipment. The present document clauses 5 and 6.2.1 [i.1], clauses 5.2.1 and 6.2.2	*This is an **existing metric** in ETSI ES 203 228 [i.1] in force.* This metric is the de-facto KPI for EE of mobile networks and beyond. BUT it does not account for contextual factors such as population density, national or state obligations (e.g. emergency services provision, rural coverage, regulatory limitations or requirements), nor does it reflect quality or availability of service.	It is one of the key metrics to plan, measure, monitor and optimize energy efficiency. BUT normalization per context and based on other factors is essential when interpreting or comparing energy efficiency metrics across different operators and geographies. It also may still fall short to cover the other purposes of networks to provide wide coverage at high availability/reliability, and/or address specific use cases of e.g. EE based service delivery optimization, when data volume is not the primary focus.	Its inverse, e.g. Wh per GB, is the most relatable to the public. But for operators, the GB per kWh is a straight forward indication to match the traffic volume with energy cost. This metric depends on the PS and CS DL/UL/both user data volume of the selected network. A clear definition of the user data volume measurement ensures ease of understanding.	Measurements are in use and are easy to measure.	For the same network and especially same traffic pattern and traffic distribution e.g. across devices/users and cells.	Not specific to this metric, rather depends on the uncertainty level especially for the energy consumption. [i.1] refer to [i.10] which specifies accuracy (e.g. ±3 % to ±5 %). Data volumes are accurately reported and precise enough within 3GPP system.	The energy proportionality is not linear and in certain contexts not fully correlated, e.g. if a capacity layer is activated at a certain traffic load threshold, then the energy consumption can jump accordingly. When it is already activated, more traffic volume does not necessarily translate directly into more energy consumption. There is also the baseline energy consumption to maintain minimum service and coverage, even at low/no traffic load.	This metric is scalable both geographically (i.e. whole network to component level) as well as based on time (i.e. full year to hourly/second).	This metric can be both standalone as well as could be multiplicative with EE metrics such as latency, coverage.

EE Metric	Added value	Relevance to use cases	Ease of understanding	Ease of measurement	Reproducibility	Reliability & accuracy	Energy proportionality	Scalability in scope and time	Standalone, composite or both
Service volume/ availability - Coverage Area $EE_{MN,CoA}$ [m^2/J] $= \frac{CoA_des_{MN}}{EC_{MN}}$ the ratio between the area covered by the MN under investigation and the energy consumption The present document, clauses 5, 6.2.1 and 6.4, [i.1], clauses 5.2.2 and 6.2.3	*This is an **existing metric** in ETSI ES 203 228 [i.1] in force.* Coverage is an essential "useful outcome" and a regulatory requirement for most mobile networks. Hence it should be covered, the question is rather which coverage-based metrics and /or factors are the most effective and representative?	With the advancement in network deployment, management and optimization, geographical area-based coverage EE metrics tend to remain constant over a given region for a larger period of time. Hence this metric may be only useful for the temporary phase and context of green field network rollout/deployment.	A very generic formula and accessible unit (e.g. square meters per kWh).	Easy and direct to measure	Reproducibility would only be relevant when comparing network deployment within similar geographical areas especially considering topology and climate.	Different geographical areas, or regions/countries may have varying mix of traffic profile, network technology and spectrum leading to different energy consumption profiles. These are not considered in this metric. It may be more representative to consider the lower bound of the network energy consumption with the lowest traffic load and energy saving optimization intended to ensure coverage & critical services.	Due to the various topologies and telecom technology in operation for a given region to derive energy proportionality is a challenge.	As coverage is an essential characteristic of mobile networks, its scalability in time and space dimension is more a matter of network rollout planning than a question of EE optimization.	For established region/country-wide network, the geographical area-based coverage is not varying significantly, hence it does not make sense as a multiplicative factor, unless networks' EE are compared with different coverage.
Service volume/quality - Latency $EE_{MN,\ L}$ [ms^{-1}/J] $= \frac{1}{T_{e2e,MN} * EC_{MN}}$ the inverse ratio of the end-to-end User Plane latency and the energy consumed by the MN. The present document, clauses 5 and 6.2.3 [i.1], clauses 5.2.3 and 6.2.4	*This is an **existing metric** in ETSI ES 203 228 [i.1] in force.* Primarily relevant for latency-focused networks.	This metric is relevant where the URLLC use cases are predominant such as industrial/enterprise consumers of mobile networks.	This metric has challenges in public networks where the uRLLC use cases are scattered. In case of private network solutions where latency together with data volume contributes to the energy consumption where this metric may play a vital role.	Needs precise latency measurements.	Might be difficult to reproduce in case of public networks. May be only relevant for subset of the network or its services (e.g. slices).	Unclear.	Energy proportionality is very likely non-linear, and it may also depend on the latency performance range of the wireless technologies in use.	Use of the metric within private network solutions such as slice, PNP, etc., and for lower time granularity gives better correlation and utility.	A multiplicative metric together with data volume.

EE Metric	Added value	Relevance to use cases	Ease of understanding	Ease of measurement	Reproducibility	Reliability & accuracy	Energy proportionality	Scalability in scope and time	Standalone, composite or both
Service volume - no of subscribers registered $EE_{MN,MMTC}[J^{-1}] = \frac{N_{MMTC}}{EC_{MN}}$ the ratio between the number of subscribers (N_{MMTC}) registered to the network, and the energy consumption. The N_{MMTC} is measured according to the definition in [i.2], clause 6.7.2.4.1. (massive Machine Type Communications: mMTC) The present document, clauses 5 and 6.2.2, [i.1], clauses 5.2.4 and 6.2.5	*This is an **existing metric** in ETSI ES 203 228 [i.1] in force.* Primarily relevant for massive MTC networks.	This metric is based on no of registered subscriber. Registered subscribers include RRC idle and RRC inactive states too. During these RRC states the subscriber network resources and hence energy usage is far less compared to RRC connected state. If there are higher number of RRC connected subscriber compared to idle and inactive subscribers, utilizing this metric would aid to derive the efficient capacity of the network in relation to the subscribers.	A very generic formula, although to the public, the differences between registered subscribers and truly connected users may not be clear and intuitive.	Easy and direct to measure	Yes, if the ratio between inactive/idle and connected users is taken into account.	The reliability and accuracy of this metric increase with a higher proportion of RRC connected subscribers compared with RRC idle and RRC inactive subscribers. Considered all registered subscribers, including the inactive and idle, may be more accurate or representative of the coverage dimension. Accuracy of the service volume aspect could be more accurate with specific tracking of the number of connected users.	Assuming a fixed RRC idle and RRC inactive an increase in RRC connected would increase the energy consumption. While with increase in RRC connected would increase energy consumption, it would have dependency on the traffic pattern and services utilized by the RRC connected subscribers. In most cases, with possibly different split or evolution of the ratio between inactive/idle and connected users, the proportionality cannot be relied upon.	This metric seems more appropriate to track more local and granular networks e.g. private wireless network covering industrial sites with massive amount of sensors and IoT devices, or massive events e.g. stadiums. On whole network level, it does bring the aspect of the number of subscribers served, which is not necessarily a direct mapping to the data traffic volume.	Together with traffic pattern would aid to optimize the energy efficient configuration of the network and no of subscribers.
Service quality - average user throughput (user throughput DL/UL/jointly (DL RAN UE throughput) / 〚EC〛_(per gNB) or UL RAN UE throughput or DL+UL UE throughput As single KPI [i.2] or as a quality factor (the present document, clause 7.2) The present document, clause 6.3.1, 7.2 and 7.4 Table 4	*This is a **NEW metric** which is not present in ETSI ES 203 228 [i.1] in force.* The service quality dimension of the overall average user throughput of a network may be a relevant dimension to include. The metric approach of normalizing the EE per BS, brings the capability to assess networks of different size and different deployment phase. This could also be considered for other EE dimensions.	Gives insights on the energy consumption to meet the service quality for service offered. It is also related to the monetization of the network considering the context that mobile operators would have different subscription plans with different level of e.g. maximum user throughput. BUT - how to normalize the impact of increased users - how to normalize the impact of Bandwidth/Spectrum.	The metric is quite relatable for end-user as the throughput is widely used and tracked across the ICT industry. However, as a standalone metric, it does not carry the information on the service volume e.g. how much traffic, how many users, hence for network energy efficiency assessment, its use requires the overall context of the network. In other words, this metric taken alone could be confusing and leading to misunderstandings.	Performance counters are available. BUT Energy Consumption is normally measured per multi-RAT BS (all Technologies) while Throughput counters are often per RAT/Technology.	Many variables affect the user throughput; hence it may be difficult to ensure similar circumstances and conditions. There is likely a strong interdependency with how the traffic volume evolves, as well as with the user application and usage patterns (rebound effect or not).	The reliability and accuracy of network measurement of user throughput, as field measurements often indicate a difference between the user throughput reported by networks, compared to what is measured on the device side.	The availability of spectrum, its usage for more advanced generation of wireless networks and the penetration level of more capable devices both have a major impact on how the **network** energy consumption may or may not be correlated to the average **user** throughput. For instance, 5G can deliver more (than previous generations) for the same amount of Energy as it has a Higer value of Throughput per Joule of Energy. But this metric is not very reliable as User Throughput is impacted by Connected users.	Scalable in both scope and time.	As a standalone metric, it does not carry the information on the service volume e.g. how much traffic, how many users, hence for network energy efficiency assessment, its use requires the overall context of the network.

EE Metric	Added value	Relevance to use cases	Ease of understanding	Ease of measurement	Reproducibility	Reliability & accuracy	Energy proportionality	Scalability in scope and time	Standalone, composite or both
Service quality/availability - Coverage quality CoA_Qdes / Ec where CoA_Qdes = (1 - RRC setup failure ratio) * (1 - RAB setup failure ratio) * (1 - RAB release failure ratio) ; [i.1], clause 6.2.3.3. The present document, Annex B.	*This is an **existing metric** in ETSI ES 203 228 [i.1] in force, however only used to calculate Coverage Area (see above)* Coverage is an essential "useful outcome" and a regulatory requirement for most mobile networks. Hence it should be covered, the question is rather which coverage-based metrics and /or factors are the most effective and representative?	This metric considers the coverage quality which enables to improve the quality of coverage in an energy efficient way. It represents the quality/ reliability/availability of the coverage, but not necessarily its extent, so this metric would be more adequate to the context when a given network is already extended and dense enough to cover its expected geographical coverage.	Based on standard defined KPIs	Easy to measure	Yes, although as it depends on radio conditions and interference levels, there could be a lot of different changing variables that would make the reproduction of similar situation more difficult.	It does not represent or consider the lack of coverage i.e. devices/users that needs a connection in remote areas. If such EE metric would consider the interconnection of public wireless networks and NTN, both the quality and extent of coverage could be put in perspective against energy consumption.	A denser network should help to address the coverage gaps, in that case there should be correlation. However, coverage is very much dependent on which spectrum is available and widely used (for instance lower FDD frequencies can reach better the devices through walls and longer distance/cell size.	Scalable in both scope and time.	This metric can be both standalone as well as could be multiplicative with other EE metrics.
Service quality - Coverage quality Calculated via RSRP reports. The present document, clauses 7.2 and A.3.	*This is a **NEW metric** which is not present in ETSI ES 203 228 [i.1] in force.* The RSRP-based approach is complementary to the existing CoA_Qdes (see above and clause A.3).	Only introduced as an adjustment factor, i.e.. not intended as standalone metric.	RSRP (Reference Signal Received Power) is the average received power of the reference signals. For coverage, RSRP is used as the primary "signal strength" metric higher (less negative) RSRP = stronger signal = better coverage potential, all else equal.	RSRP reports are readily available as standardized RAN counters in any vendor eNB/gNB.	The amount of RF related variables are quite high: bands, frequencies, interference levels and sources, multi-RAT configuration, 5G NSA or SA, 4G-5G mobility configuration, number of sites, etc.	In both LTE and NR, coverage quality is largely interpreted through RSRP distributions and the thresholds used in measurement/handover events.	Not a given as it depends on the maturity of the coverage optimization. Coverage quality issues measured via RSRP reports could be addressed by tilt/power/height changes, while only the deployment of additional sites is the additional approach that would translate into higher energy consumption.	This is a radio related quality criterion, so it can scale from BS to whole RAN.	Only introduced it as an adjustment factor, hence composite only.
Service quality - Reliability The present document, Annex B. [i.1] clause 6.2.2 Measurement of capacity e.g. for CS call success rate, which is based on the "dropping rate"	*This is an **existing metric** ETSI ES 203 228 [i.1] in force.*	The present document mentions it as an adjustment factor for uRLLC. The accessibility (capability to make an emergency call, a normal call, whether CS voice, or VoLTE/VoNR) is paramount to coverage.	Call success rate and call drop rate are easy to understand, *but reliability is not only about calls, when considering the major share of IP-based traffic e.g. internet page loading, data connection reliability, etc.*	It will only cover the call success rate that the operators have visibility for, so not the calls done via OTT services e.g. WhatsApp, which can represent significant share of the total calls.	Depends on the usage and service patterns, and is dependent on many variables.	It will only cover the call success rate that the operators have visibility for, so not the calls done via OTT services e.g. WhatsApp, which can represent significant share of the total calls.	Not a given as it depends on the maturity of the coverage optimization as well as the end to end testing and compatibility support for VoLTE and VoNR calls.	Finer granularity of its usage may prove problematic if the volume of real traditional calls is low.	Composite only.

EE Metric	Added value	Relevance to use cases	Ease of understanding	Ease of measurement	Reproducibility	Reliability & accuracy	Energy proportionality	Scalability in scope and time	Standalone, composite or both
Service availability - RAN Availability EE (Radio access network availability) / 〖EC〗_(per cell) The present document, clauses 6.4, 6.5, and 7.4 Table 5 [i.5], clause 5.6.3.2	*This is a **NEW metric** which is not present in ETSI ES 203 228 [i.1] in force.* Addresses one aspect of network/service availability dimension. Added value is unclear.	Maybe only relevant for use cases when RAN cell availability is not already optimized to high values (e.g. 99%). AND it might not bring much added value compared to DV EE, as there is no traffic for the cells which are not available.	Unclear.it might not bring much added value compared to DV EE, as there is no traffic for the cells which are not available.	Based on existing 3GPP KPIs, but only for 5G gNB sites.	Unclear how this metric allows to differentiate between cells down due to network issues and cell off for energy saving?	Unclear, it would need field network data-point for review.	There should be direct correlation between energy consumption and cell availability as the radio modules are the biggest contributors to the overall network energy consumption.	Only focused on 5G RAT and network. Also likely only relevant for longer time span considering its definition and scope.	Proposed, but not introduced in 3GPP as standalone KPI [i.5].
Service availability - Cell availability EE (Cell Availability KPI) / 〖EC〗_(per cell) The present document, clauses 6.4, 6.5 and 7.4, Table 5 [i.2], clause 6.7.5.3	Same comments as RAN Availability EE.	Same comments as RAN Availability EE.	Same comments as RAN Availability EE.	Same comments as RAN Availability EE.	Same comments as RAN Availability EE.	Same comments as RAN Availability EE.	Same comments as RAN Availability EE.	Same comments as RAN Availability EE.	Introduced in 3GPP as standalone KPI [i.2].

8.3 Qualitative analysis of multi-dimensional metric proposals

While the previous clause focused on evaluating the new or revised EE metrics, i.e. the specific performance metrics they are based on, this clause focus on qualitatively analysing the overall multi-dimensional metric proposals - metric based on adjustment factor(s), multiplicative metric, and the approach with separate metrics per performance dimension.

The qualitative analysis will discuss the metric proposals from four aspects:

1) how easy it is to understand the metric;
2) how difficult it is to construct the metric;
3) single vs multiple metrics; and
4) applicability of the metric.

Easy to understand the metric: This criterion assesses how easy it is to understand the metric. This is important when communicating the metric, but also when it is used in e.g. optimization algorithms.

Typically, a single-dimensional metric is easy to understand as it gives a straight-forward relation between a specific type of performance and energy consumption. It measures the energy cost for the specific performance. Similarly, also a multi-dimensional metric using a unitless adjustment factor may be easy to understand, as it simply scales the single-dimensional base metric based on the performance metric reflected by the adjustment factor.

However, a multiplicative metric can be difficult to understand, as it mixes units of different performance metrics and relates them to the energy consumption at the same time. It may then be difficult to understand how the metric behaves.

A big challenge with a multi-dimensional metric that incorporates and combines service volume, service quality, service availability and potentially other aspects is that it risks to hide important nuance. Hence, for those who seek energy efficiency improvements - a top-line multi-dimensional metric would catch one's attention - but to drive energy efficiency improvements, one would still need to refer to the single-dimension metrics to better understand what is driving the results.

Difficulty to construct the metric: This criterion assesses how difficult it is to construct the metric. A single-dimensional metric is straightforward to construct, typically it is to identify the performance metric to base the metric on, and divide by the energy consumption.

A multiplicative metric is one step up in difficulty, as it requires identifying two or more performance metrics to include in the metric, but once that is done the construction is straight-forward.

Introducing a unitless adjustment factor to construct the multi-dimensional metric makes the process more difficult. Not only is it necessary to identify the performance metric(s) to base the adjustment factor(s) on, but it is also necessary to define the targeted value of this metric that is used to normalize the measured value in the calculation of the adjustment factor. This requires a competent user of the metric.

Another challenge when constructing a multi-dimensional metric is how one can fairly combine and balance the multiple dimensions into a single metric. Different telcos are likely to take different views on the weighting for the factors to be used, reflecting their specific networks and priorities.

Single vs multiple metrics: A single metric capturing multiple dimensions is easier to use in e.g. optimization algorithms as well as in reporting, while multiple single-dimensional metrics may give a wider picture of the network energy performance.

Applicability of the metric: This criterion assesses how flexible the metric is in terms of use. A single-dimensional metric is rather inflexible, while a metric with adjustment factor can be used in several ways; without limitation, with limitation, with inverted adjustment factor to punish non-optimal resource usage (see clause 7.2.2), etc. On the other hand, the latter then also requires a competent user of the metric.

9 Conclusions and recommendations

The scope of the present document has been to study existing network energy efficiency metrics and outline proposals on how network energy efficiency metrics capturing multiple performance dimensions (service volume, service quality, service availability, etc.) can be formulated. These can be combinations of existing network energy efficiency metrics, but also new proposals capturing performance dimensions not considered today. The focus has been on network energy efficiency metrics to be used for monitoring and/or optimization of live networks, even though metrics for reporting and related implications have been also covered.

The background is that existing network energy efficiency metrics are typically only based on the amount of service provided, e.g. transported data volume, coverage area, or number of connections. Other dimensions of useful work, e.g. the quality of the service provided, is seldom reflected, and it is also not always clear if/how these metrics can be combined to give a more complete picture of the energy efficiency of a network.

In general, three performance dimensions have been considered, and related performance indicators identified and discussed:

- **Service volume:** Data volume (capacity sites), coverage area (coverage sites), number of users/connections.
- **Service quality:** Minimum or Average user throughput, Minimum coverage (e.g. for basic and critical services), connection accessibility and reliability, Average connection/data transfer latency.
- **Service availability:** Ensuring capability and quality are there.

To be a good performance indicator to be included in the analysis, the performance indicator should be related to energy consumption directly, i.e. when this performance factor decreases, energy consumption decreases; or when this performance factor increases, energy consumption increases (negative correlation). Furthermore, it is also desirable that the performance factor remains relevant over time, over network areas, for different scenarios and service type e.g. eMBB, FWA, network slices/private networks, massive mMTC, etc.

The performance indicators that currently are included in ETSI ES 203 228 [i.1] and there are energy-efficiency metrics based upon are data volume (in bits), coverage area (in m^2), latency (in ms), and number of registered subscribers. Hence, the focus of the present document has been on what additional performance indicators should be considered, and how multi-dimensional metrics (taking several performance indicators into account) could be formulated.

The analysis concludes that it is desirable that in addition to the existing mostly service-volume based EE metrics, also service quality is added into the EE assessment. In particular, service quality based on (average) user throughput may be valuable to consider in addition to data volume when assessing mobile network data energy efficiency, something that has also been concluded by 3GPP TR 28.880 [i.5] ETSI TS 128 554 [i.2]. It is also desirable to extend the current mobile network coverage energy efficiency analysis in ETSI ES 203 228 [i.1] to consider differences in coverage quality in addition to coverage availability, see Annex B. Furthermore, there might also be a need to investigate a better representation of coverage delivery and quality to extend the usefulness of mobile network coverage energy efficiency assessment beyond network rollout/greenfield deployment, however, this is left for further study.

Whether also service availability should be considered is less obvious, as the analysis conducted shows limited value to add such metrics into the assessment.

Three multi-dimensional metric proposals, i.e. structures for considering multiple performance dimensions in the EE assessment, have been defined and assessed in the present document:

- **Metric based on adjustment factor(s):** A base metric, typically the existing service volume based EE metrics already defined in ETSI ES 203 228 [i.1], is extended (i.e. multiplied) by a dimension-less adjustment factor. The idea is that the adjustment factor captures another performance dimension and scales the base metric accordingly.
- **Multiplicative metric:** In this case, two performance indicators are multiplied and considered in relation to the network energy consumption.
- **Separate metrics per performance dimension:** Multiple single-dimensional energy efficiency metrics represent the different performance dimensions in the network energy efficiency evaluation.

Of these, the second proposal - multiplicative metric - has been assessed as least attractive. It can be difficult to understand, as it mixes units of different performance metrics and relates them to the energy consumption at the same time. It may then be difficult to understand how the metric behaves, and what it really tells. The other two have their pros and cons depending on the use case. Using separate metrics per performance dimension is beneficial for a more nuanced assessment, as it reveals all details of what is driving the results. However, it may make an optimization algorithm more complex with more variables to consider. A metric based on adjustment factors may hide nuances, but with just one or two performance indicators reflected this risk is reduced. Furthermore, if it is constructed in a sound way it should be easy to understand, e.g. with service volume scaled by a factor based on service quality. It may in particular be useful for network optimization, as exemplified in Annex C.

For reporting, it is recommended to keep the existing single dimensional metrics, e.g. mobile network data energy efficiency, as they have been used for long time, are well-known, and also requested in some reporting frameworks, see e.g. [i.11].

Furthermore, the analysis also confirms that contextual factors such as geographical conditions, population density, topology, climate zones, and also regulatory obligations on coverage and service quality, may have a significant impact on the energy efficiency assessment and its' results. Therefore, this again emphasizes what is stated in [i.1] that the network level energy efficiency analysis and metrics should not be used for benchmarking of different networks (unless the contextual factors are the same or compensated for), rather the use case is for assessing improvements over time in a specific network and optimization of the network.

Annex A:
Required measurements for adjustment factors

A.1 Introduction

In clause 7.2 above, two quality adjustment factors for the eMBB use case were discussed, $Factor_{throughput}$, and $Factor_{coverage}$. They were defined as follows:

$$Factor_{throughput} = \frac{Average\ downlink\ user\ throughput}{Average\ target\ downlink\ user\ throughput}$$

and

$$Factor_{coverage} = \frac{Cell\ edge\ user\ throughput}{Target\ cell\ edge\ user\ throughput}$$

A.2 Throughput measurement

The adjustment factor $Factor_{throughput}$ is based on the average downlink user throughput. The average downlink throughput is easy to calculate based on standardized counters available in the Operation and Maintenance (O&M) system, as described in clause 6.3.1 in ETSI TS 132 450 [i.4] for LTE and clause 6.3.6 in ETSI TS 128 554 [i.2] for 5G-NR.

To achieve a DL throughput measurement that is independent of file size and gives a relevant result it is important to remove the volume and time when the resource on the radio interface is not fully utilized. (Successful transmission, buffer empty in Figure A.1). To achieve a throughput measurement that is independent of bursty traffic pattern, it is important to make sure that idle gaps between incoming data is not included in the measurements. That is done by considering each burst of data as one sample.

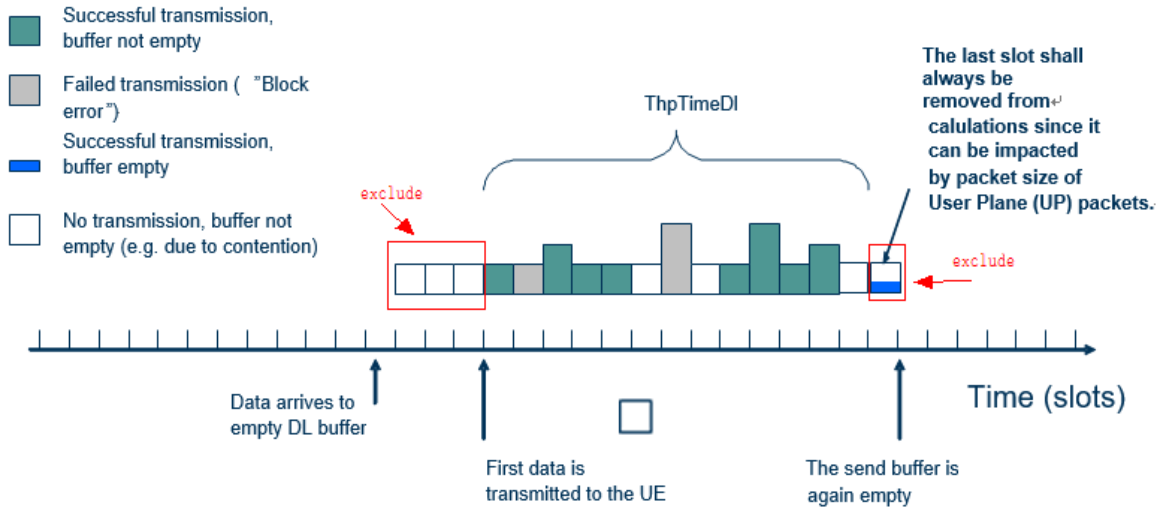


Figure A.1: Illustration of calculation of average DL throughput (see ETSI TS 128 554 [i.2] and ETSI TS 132 450 [i.4])

A summary of the KPI calculation based on counters is provided in Table A.1.

Table A.1: Summary of KPI calculation based on ETSI TS 128 554 [i.2] and ETSI TS 132 450 [i.4]

RAT	Counter	Description	KPI calculation	KPI
LTE	L.Thrp.bits.DL	Total downlink PDCP-layer traffic volume in a cell	(L.Thrp.bits.DL - L.Thrp.bits.DL.LastTTI)/L.Thrp.Time.DL.RmvLastTTI	User Downlink Average Throughput (4G)
	L.Thrp.bits.DL.LastTTI	Downlink PDCP-layer traffic volume sent in the last TTIs before the buffer is empty		
	L.Thrp.Time.DL.RmvLastTTI	Data transmission duration excluding the last TTIs before the downlink buffer is empty		
NR	N.ThpVol.DL	Total volume of downlink data sent at the RLC layer in a cell	(N.ThpVol.DL - N.ThpVol.DL.LastSlot)/N.ThpTime.DL.RmvLastSlot	User Downlink Average Throughput (5G)
	N.ThpVol.DL.LastSlot	Downlink volume of RLC SDUs sent in the last timeslot during which the buffer becomes empty		
	N.ThpTime.DL.RmvLastSlot	Data transmission duration excluding the last timeslot during which the downlink buffer becomes empty		

A.3 Coverage quality measurement

The adjustment factor $Factor_{coverage}$ was in clause 7.2 (and in the equation above) based on cell edge user throughput. However, cell edge user throughput cannot easily be calculated from standardized counters as the average downlink user throughput discussed above. As mentioned in clause 7.2, an alternative is to use RSRP reports to represent the coverage quality (e.g. the amount of "good" RSRP reports in relation to the total amount of RSRP reports). Due to the difficulties to establish cell edge user throughput measurements based on standardized counters, and the fact that RSRP reports are readily available in any vendor eNB/gNB, it is proposed to define $Factor_{coverage}$ as:

$$Factor_{coverage} = \frac{Coverage\ quality}{Target\ average\ coverage\ quality}$$

where the $Coverage\ quality$ can be defined as the amount of RSRP reports above a certain threshold divided by the total amount of RSRP reports. This calculation can be simplified by utilizing counters that measure the distribution of RSRP reports within a cell with respect to pre-defined ranges, e.g. 0 to n, where the worst reports are categorized in range 0 and the best reports are categorized in range n. KPI calculation can then be carried out as in Table A.2, in which index k is chosen to represent the threshold range for sufficiently good RSRP reports.

Table A.2: Summary of coverage quality KPI calculation

RAT	Counter	Description	KPI calculation	KPI
LTE	L.MeasRpts.RSRP.Index0~n	Number of samples with the measured reports RSRP ranging within index 0~n	$\frac{\sum_{j=k}^{n} \mathrm{L.MeasRpts.RSRP.Index}_j}{\sum_{j=0}^{n} \mathrm{L.MeasRpts.RSRP.Index}_j}$	Coverage Quality (4G)
NR	N.MeasRpts.RSRP.Index0~n	Number of samples with the measured reports RSRP ranging within index 0~n	$\frac{\sum_{j=k}^{n} \mathrm{N.MeasRpts.RSRP.Index}_j}{\sum_{j=0}^{n} \mathrm{N.MeasRpts.RSRP.Index}_j}$	Coverage Quality (5G)

This means that in the KPI calculation, the numerator includes all RSRP samples greater than a given threshold (e.g. -110 dBm) and the denominator includes all the RSRP samples. Hence the KPI takes a value between 0 and 1, the higher the better coverage quality.

Annex B:
Comparison of existing and proposed quality adjustment factors

B.1 Introduction

The objective of the present document, "Study of multi-dimensional network energy efficiency metrics", is to investigate how in particular differences in service quality can be reflected in the service-volume based energy efficiency metrics that typically are used today. The motivation being that using metrics only based on service volume may lead to incorrect conclusions, and also that energy efficiency optimizing methods may be mis-lead if only the service volume is considered. One proposal to capture the service quality dimension, introduced in clause 7.2, is to introduce adjustment factors and multiply the service volume based metric with them.

However, ETSI ES 203 228 [i.1] already contain some factors and methods that are supposed to reflect quality, e.g. there is a factor for coverage quality and a similar factor for circuit-switched data volume (typically voice traffic). In addition, it can also be observed that the measured packet data volume in ETSI ES 203 228 [i.1] considers QoS classes.

This annex review and discuss these service quality factors, and try to explain the fundamental difference between the already existing factors for some services in ETSI ES 203 228 [i.1] and the ones being proposed in clause 7.2.

B.2 Adjustment factors in the present document

For the eMBB use case, there exist two energy efficiency metrics in ETSI ES 203 228 [i.1], one based on data volume (data volume/energy consumption) and one based on coverage area (coverage area/energy consumption).

In the present document, one proposal is to take these existing volume-based metrics as starting point, and to introduce an adjustment factor to reflect the service quality dimension. The adjustment factor would be dimensionless, in order not to affect the unit of the metric. Hence, a metric taking service quality into account would look like this:

$$Energy\ Efficiency = \frac{Service\ volume}{Energy\ consumption} \times Factor_{quality}$$

The adjustment factor, $Factor_{quality}$, should represent the service quality. If data volume is considered, a reasonable service quality indicator is the user throughput. To achieve a dimensionless metric, the adjustment factor is defined as the average downlink user throughput in relation to the targeted downlink user throughput that the MNO wants to offer to its customers:

$$Factor_{quality} = \frac{Average\ downlink\ user\ throughput}{Average\ target\ downlink\ user\ throughput}$$

Hence, it describes how well the network is able to deliver the intended service quality, in this case, downlink user throughput. The average downlink throughput is easy to calculate based on standardized counters available in the operation and maintenance (O&M) system (ETSI TS 132 450 [i.4] and ETSI TS 128 554 [i.2]), see also Annex A for details.

To exemplify, take the networks introduced in Table B.1, and assume that the average targeted downlink user throughput is 10 Mbit/s. The adjustment factors (in this case, $Factor_{quality}$) and the adjusted energy efficiencies for the different networks can then be calculated. The results are summarized in Table B.1. It can be seen that network A, which is the second best of the three in terms of network data energy efficiency, but is rather poor in delivering the targeted downlink user throughput of 10 Mbit/s, becomes the worst of the three in terms of QoS-adjusted energy efficiency. Network B, which is slightly worse that network A in terms of network data energy efficiency, but good in delivering the targeted downlink user throughput, is better than network A, when the QoS-adjusted energy efficiencies are compared. Finally, network C is the best both before and after QoS adjustment. However, note that its delivered downlink user throughput is much higher than the targeted downlink user throughput. In order not to benefit from this over-provisioning, the adjustment factor is in this case limited to 1.

Table B.1: Comparison of network data energy efficiency, user throughput, and QoS-adjusted energy efficiency in different networks

Network	Network Data Energy Efficiency (GB/kWh)	Average DL User Throughput (Mbit/s)	$Factor_{quality}$	QoS-adjusted Network Data Energy Efficiency (GB/kWh)
A	6,4	6,3	0,63	4,0
B	6,2	9,6	0,96	6,0
C	6,7	20,1	1	6,7

For the network energy efficiency metric based on coverage area, it is natural to assume that the adjustment factor, $Factor_{quality}$, should represent the coverage quality. One such measure could be the cell edge throughput in relation to the targeted cell edge user throughput. However, it is difficult to measure the cell edge throughput based on counters in the O&M system, instead there is a proposal to use RSRP reports to represent the coverage quality (e.g. the amount of "good" RSRP reports in relation to the total amount of RSRP reports), see Table B.2.

Table B.2: Summary of coverage quality calculation

RAT	Counter	Description	KPI calculation	KPI
LTE	L.MeasRpts.RSRP.Index0~n	Number of samples with the measured reports RSRP ranging within index 0~n	$\frac{\sum_{j=k}^{n} \mathrm{L.MeasRpts.RSRP.Index}_j}{\sum_{j=0}^{n} \mathrm{L.MeasRpts.RSRP.Index}_j}$	Coverage Quality (4G)
NR	N.MeasRpts.RSRP.Index0~n	Number of samples with the measured reports RSRP ranging within index 0~n	$\frac{\sum_{j=k}^{n} \mathrm{N.MeasRpts.RSRP.Index}_j}{\sum_{j=0}^{n} \mathrm{N.MeasRpts.RSRP.Index}_j}$	Coverage Quality (5G)

This means that in the coverage quality calculation, the numerator includes all RSRP samples greater than a given threshold (e.g. -110 dBm) and the denominator includes all the RSRP samples. Hence the coverage quality based correction factor always takes a value between 0 and 1, the higher the better coverage quality.

B.3 Existing factors in ETSI ES 203 228

B.3.1 Introduction

Already in the current version of ETSI ES 203 228 [i.1] there exist a sort of quality adjustment factors for Circuit-Switched (CS) service volume and for the coverage area.

B.3.2 CS traffic

For CS traffic (e.g. CS voice or VoLTE), the data volume is considered as the number of minutes of communications during the time T multiplied by the data rate of the corresponding service and the call success rate. The call success rate is equal to 1 minus the sum of blocking and dropping rates, i.e.:

$$Call\ Success\ Rate = (1 - dropping\ rate) \times 100\ [\%]$$

The dropping includes the intra-cell call failure (rate of dropping calls due to all the causes not related to handover) and the handover failure:

$$1 - \text{dropping rate} = (1 - \text{intracell failure rate})(1 - \text{handover failure rate})$$

B.3.3 Coverage area

Similarly, also the designated coverage area is to be multiplied by a sort of quality factor in ETSI ES 203 228 [i.1].

The coverage quality factor for a base station is based on network failure reports of the UE. The coverage quality factor is measured based on coverage failures reported by the appropriate network counters:

CoA_Qdes = 1 - "percentage of users/sessions with coverage failure"

The following indicators are used to calculate the coverage failure:

1) RRC setup failure ratio (Call setup failure ratio).

2) RAB setup failure ratio (UE-BS radio interface failure).

3) RAB release failure ratio (UE-BS radio interface failure).

A further factor which can indicate a coverage issue is the handover drop ratio. However, a handover drop can have multiple reasons (cell overload, UE speed, etc.). Furthermore, the handover drop rate depends on the network structure (number of neighbour cells). Its calculation requires several additional network parameters and complicates the data collection and analysis significantly. This factor is therefore omitted.

The coverage quality factor for a site is defined as follows:

CoA_Qdes = (1 - RRC setup failure ratio) (1 - RAB setup failure ratio) (1 - RAB release failure ratio)

The needed parameters are specified by ETSI standards and the results can be obtained from the network management and supervision.

B.4 Observations and discussion

The existing quality factors for CS traffic and coverage area in ETSI ES 203 228 [i.1] are **focused on the situation where the service cease to exist**, i.e. when the CS call drops or the coverage is so bad that it is impossible to setup a connection. It is obvious to not include these situations when calculating the service volume (i.e. CS traffic volume and coverage area).

Furthermore, the packet data volume in ETSI ES 203 228 [i.1] is calculated using counters taking the QoS Class Identifier (QCI) into account. However, the effective result of this is also focused on the situation where the service cease to exist (i.e. when the required minimum QoS for a service is not fulfilled). It should also be noted that QCI only applies to certain types of services, and not to best effort services such as web browsing (better QoS means web pages load faster) or video streaming (where a better QoS results in e.g. higher resolution).

To conclude, the already existing factors in ETSI ES 203 228 [i.1] are only focused on the situation when the service cease to exist (by removing these situations when calculating the service volume), **but are not able to capture differences in service quality** which is the objective of the metrics being studied in the present document.

Referring to Figure B.1, the already existing factors in ETSI ES 203 228 [i.1] cut away the red part where there is no service, but can after that only distinguish in the service volume (horizontal) dimension, resulting in that A is better then B despite the fact that B delivers slightly better quality than the targeted service quality while A is far below the targeted service quality. The adjustment factors in the present document are able to capture the difference in service quality (the vertical dimension) and "punish" A for its poor service quality below the targeted service quality.

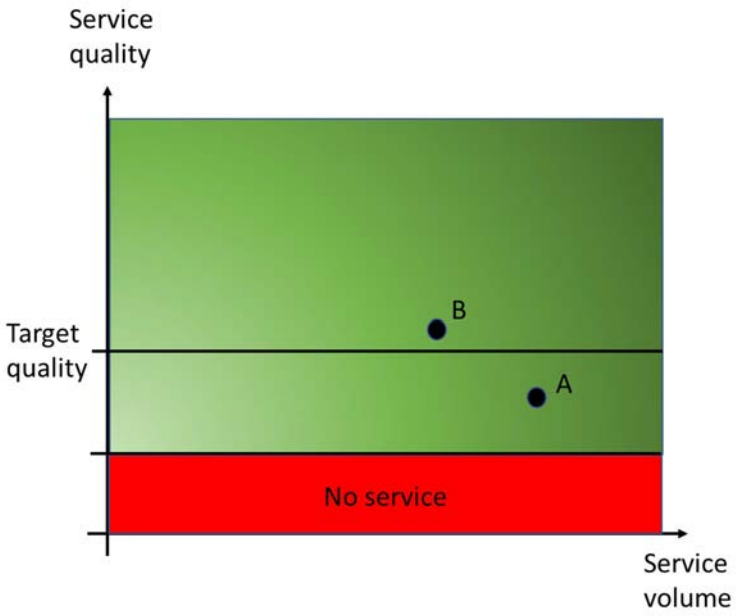


Figure B.1: Illustrative diagram of service volume vs service quality and what existing adjustment factors and new adjustment factors capture

Annex C:
Example application of metric based on adjustment factor

C.1 Introduction

In this annex, an example of how a multi-dimensional energy efficiency metric is used to analyse and optimize a mobile network in a European metropolitan area is provided. The metric considered is the QoS-adjusted metric introduced in clause 7.2, and also shown here:

$$\text{Energy Efficiency} = \frac{Data\ volume}{Energy\ consumption} \times Factor_{quality}$$

with

$$Factor_{quality} = \frac{Average\ downlink\ user\ throughput}{Average\ target\ downlink\ user\ throughput}$$

C.2 Methodology

Cell-level statistics is collected from all sites in the area under study in order to derive the average downlink data volume and user throughput per site as well as its energy consumption.

Once these statistics is available, the QoS-adjusted energy efficiency metric is computed per site.

The QoS-adjusted energy efficiency metric per site is plotted versus site traffic, and a benchmark curve is fitted to the data points (note that this curve over time need to be updated for future optimization exercises to accommodate for traffic and network changes).

With the generated plot and the benchmark curve, low energy efficiency sites (and root cause) can be identified by comparing the QoS-adjusted energy efficiency metric of each site with the fitted network energy efficiency benchmark.

Energy efficiency improving actions targeted towards the different root causes are carried out on the low energy efficiency sites.

New cell-level statistics is collected, analysed, and improvements calculated.

C.3 Reference implementation

Cell-level statistics were collected from a large number of sites in a European metropolitan area, and the QoS-adjusted energy efficiency metric was computed per site, in this case using 20 Mbps as average target downlink user throughput.

Figure C.1 illustrates the results of this process, where the QoS-adjusted energy efficiency metric is plotted versus site traffic. Every orange dot represents one site, and the green lines represent current and future energy efficiency benchmark curves.



Figure C.1: QoS-adjusted energy efficiency plotted versus traffic

Also shown in Figure C.1 are the identified low energy efficiency sites and proposed actions to address the root causes and improve the QoS-adjusted energy efficiency. In summary, these can be explained as follows:

- Type I: Low traffic and high downlink user throughput, where network energy saving solutions are not fully utilized/activated.
- Type II: High traffic and low downlink user throughput, where energy saving solutions are fully utilized, and the hardware generation is old.
- Type III: High traffic and high downlink user throughput, where energy saving solutions are fully utilized, and the hardware generation is new.

For Type I sites, the QoS-adjusted energy efficiency metric can be improved by activating energy saving solutions more aggressively. By doing this, the energy consumption is reduced which improve the QoS-adjusted energy efficiency metric. Probably the downlink user throughput is decreased, but the adjustment factor is not affected as it is capped to 1.

For Type II sites, the QoS-adjusted energy efficiency metric can be improved by hardware refresh. This will decrease the energy consumption as new hardware generations typically are more efficient, and probably also improve the downlink user throughput. Both these attributes contributes to improve the QoS-adjusted energy efficiency metric.

For Type III sites, the QoS-adjusted energy efficiency metric can be improved by migrating traffic towards newer RAT generations that are more energy efficient, e.g. 5G.

C.4 Optimization performance

In this clause, it is shown how the mentioned actions for the different site types indicated earlier contributed to network energy efficiency optimization in a trial carried out on the sites presented in Figure C.1. Since the traffic mitigation from 2G/3G/4G to 5G is more complex to optimize and the gains more complex to quantify, the focus is on Type I and Type II sites in the analysis.

For Type I sites, and to quantify the gains brought by the network energy saving solution optimization, two phases were conducted in the trial. During the first phase, in order to estimate the baseline network energy consumption, all the network energy saving solutions were deactivated while taking measurements. In the second phase, the network energy saving solutions were activated and tuned for the measurement campaign.

Figure C.2 shows the network energy consumption before and after optimization. It can be seen that the energy saving solution activation and optimization at the Type I sites saves ~1 500 kWh per day. Furthermore, Figure C.3 shows how these network energy consumption reductions are achieved while maintaining stable Key Performance Indicators (KPIs) related to network performance and user service quality. As a result, the QoS-adjusted energy efficiency metric in the trial area was improved by 17,5 %, as can be seen in Figure C.4.

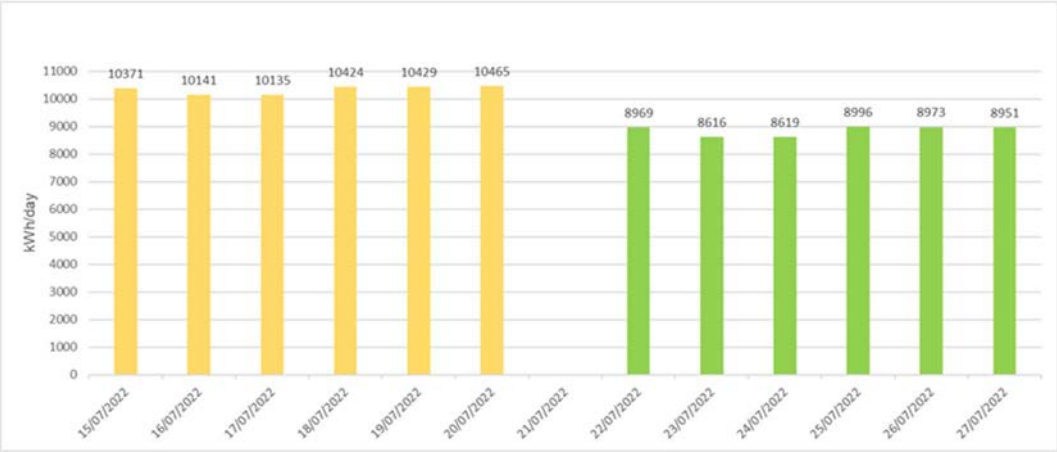


Figure C.2: Network energy consumption before (yellow colour) and after (green colour) Type I site optimization

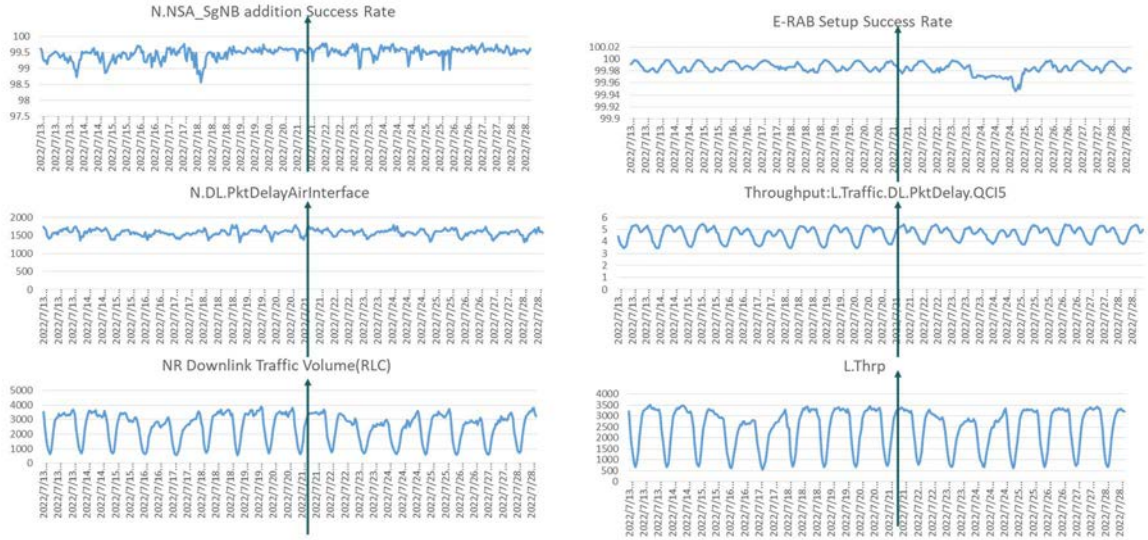


Figure C.3: LTE and NR network performance KPIs kept stable

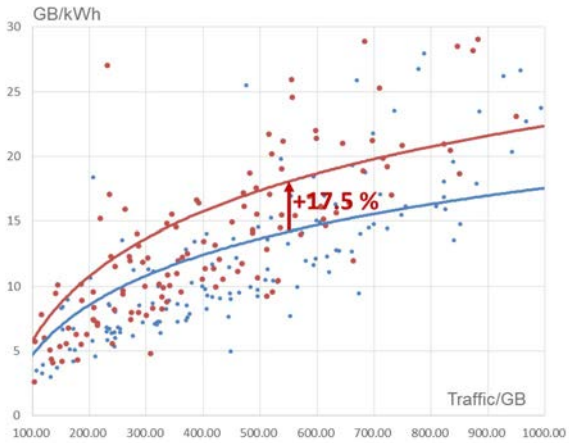


Figure C.4: QoS-adjusted energy efficiency metric gain due to Type I site optimization

For Type II sites, hardware refresh was applied. In the trial, two specific sites were selected to quantify the gains obtained by the hardware refresh. The results of the analysis, shown in Figure C.5 and Figure C.6, indicate that the hardware refresh not only resulted in 35 % to 40 % energy saving gain per site (Figure C.5), but also an improvement in user service quality as reflected in the QoS-adjusted energy efficiency metric (Figure C.6).

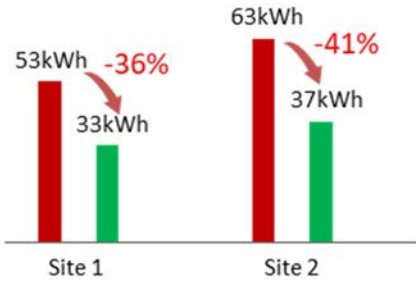


Figure C.5: Energy saving due to hardware refresh of Type II sites



Figure C.6: QoS-adjusted energy efficiency metric gain due to hardware refresh of Type II sites

History

Version	Date	Status
V1.1.1	August 2026	Publication